EDUPULSE

Complete speaker notes

Twelve-minute presentation and live application guide

CSE Year One | 70 students | 11 aligned sections

Exact spoken script, slide cues, practical clicks, panel interactions, answer keys, AI explanation, and Q&A.

PRIVATE PRESENTER COPY

Keep this PDF off the projected display. It contains answer keys and rehearsal directions. The core presentation runs locally on Windows with cloud generation disabled.

Presentation target: 12 minutes. Reserve about 3 minutes for questions.

This script presents a Year One CSE class of 70 students through four different support needs. Target **12 minutes**, leaving about **3 minutes for questions** in a 15-minute slot. The script contains approximately 1,086 spoken words; the remaining time is for navigation, reading code, and panel participation. Speak at a relaxed pace and rehearse once with a timer. The words under “Say” are spoken; screen directions and the answer key are private.

Presentation switching guide

Slides 1 and 2: explain the situation and introduce the four student paths.

Live application for sections 3 through 9: Asha evidence > lesson > retest > saved progress > Compare students with Rahul > Ishaan > Omar. Slides 3-9 are aligned backups; you do not need to switch back for every click.

Slides 10 and 11: return to the deck for architecture, the AI role, and the closing. In PowerPoint Slide Show, type **10** and press **Enter** to jump to slide 10. Use **Alt+Tab** to switch between PowerPoint and the prepared browser.

Open the answer key only on a private laptop display or printed page. Do not project the keys with the application.

Private preparation checklist

- Before beginning, disclose once: **“For this presentation I am using generated student records. The answers we submit now will be saved by the running system.”** Classroom screens need not repeat that disclosure.
- The opening is a hypothetical classroom situation. Do not present it as an incident that actually happened to you.
- Open the application with Start-EduPulse.cmd. Keep the overview projected; keep these notes on another device or an unprojected window.
- Check that the class count is 70 and that Asha, Rahul, Ishaan, and Omar are searchable. The private rehearsal state should leave Asha's nested-loop estimate at 20%, Ishaan unassessed, and Omar's review available. Do not reset data during the panel presentation.
- Keep live generation off. No API key is required. Do not upgrade an account, add billing, or buy credits.
- Confirm the event's permission for AI-assisted development and APIs. Explain the libraries and assistance used honestly.
- Use one complete Asha journey. For Rahul, show the distinct lesson; for Ishaan, show the unassessed state; for Omar, show mentor review. There is no need to complete four assessments.
- Never rehearse answers on the same prepared presentation state immediately before presenting. Completed assessments reopen as results to prevent repeated scoring. Use a separately prepared rehearsal state or have the original class restored before the slot.
- Avoid changing browser zoom mid-presentation. Check the projected text from the back of the room and close unrelated tabs.
- If you have already changed one student's records, describe the values on screen. The script's expected values depend on a fresh baseline.

0:00-1:00 A simple classroom situation

Slide cue: 1 / Problem and classroom situation / PROJECT THE SLIDE

Say:

"Good morning. Imagine a first-year Computer Science class with seventy students. The teacher finishes a lesson on loops and asks, "Is everyone clear?" A few students say yes. Some stay quiet. The class moves on. A week later, an assessment shows that several students were struggling with different parts of the lesson. Now the teacher has to work backwards: who needs help, what exactly is confusing them, and what should happen next? I used that simple, hypothetical classroom situation as my starting point. And that is how I came up with this idea: EduPulse, a system that helps faculty turn a learning gap into a clear next action."

Exact practical actions: Stay on SLIDE 1 while telling the story. Keep the live application ready in another window. This is a hypothetical situation, not a claimed personal incident. Do not read the speaker notes from the projected screen.

1:00-1:50 Introduce the class

Slide cue: 2 / One class, four support paths / PROJECT THE SLIDE

Say:

"This is the workspace for CSE, Year One. We can move from the whole class to one student, inspect a concept, open a learning plan, and return to see what changed. The important point is that seventy students do not need seventy copies of the same advice. Today I will show four situations: Asha needs help with nested loops; Rahul needs a different starting point; Ishaan has no assessment evidence yet; and Omar needs a mentor conversation. Together, they show how the same system can support different decisions."

Exact practical actions: Advance to SLIDE 2. Point to the 70-student class and the four named paths. At the end, switch to the live browser with Alt+Tab. The following seven sections are performed in the application; slides 3-9 are visual backups.

1:50-3:00 Let the panel choose the evidence

Slide cue: 3 / Asha and concept evidence / LIVE APPLICATION; slide is a backup

Say:

“Let us begin with Asha. Before I open her learning plan, may I ask you one question? If a student attends most classes and submits most assignments, is that enough to decide which topic she understands? I would still want to see her answers. Here, attendance and assignment completion provide context, while the concept scores show where to investigate. Asha's clearest gap is nested loops. The support priority also shows its ingredients, so a teacher can inspect the reasoning. It is a transparent prioritization rule. The purpose is to direct attention toward useful evidence and a practical next step.”

Exact practical actions: LIVE APP: click Students in the sidebar. In Search name or ID, type Asha. Click Asha Raman. Pause 10-15 seconds for the panel response. Show the three concept bars, then scroll to Why this support priority? Explain the values actually displayed. Generated baseline records do not include inspectable historical question text. SLIDE 3 is the backup.

3:00-4:05 Give the right lesson

Slide cue: 4 / Targeted lesson / LIVE APPLICATION; slide is a backup

Say:

“I will open Asha's learning plan. The lesson concentrates on one idea: each outer-loop iteration starts another complete run of the inner loop. The explanation is broken into small steps, followed by a worked example. The teaching reference is visible, and the practice questions give feedback. That gives the student a way to work on the gap before taking another check. In the current configuration, these are reviewed local lessons. Optional Gemini generation can rephrase approved material, but it is switched off today. The useful workflow still works. I am moving through this lesson quickly to show the system; a real learner would take the time they need.”

Exact practical actions: LIVE APP: on Asha’s profile, click Open learning plan. Show Let’s break it down, A worked example, and OUTPUT. Click Where this lesson comes from to expand the source. Scroll to I’ve reviewed the lesson and click it. Point to Practice with feedback. Then click Take independent retest. Do not claim the few seconds on screen constitute a completed learning intervention. SLIDE 4 is the backup.

4:05-6:05 Ask the panel to take part

Slide cue: 5 / Live assessment / LIVE APPLICATION; slide is a backup

Say:

“Now we check what the learner can do with different questions. Would one of you choose the first answer? There are three outer iterations and four inner iterations. How many times does the inner statement run? Please choose the option you would submit. I will enter the answer you choose. The response is saved, and feedback stays hidden until the check is complete so it does not teach the remaining answers. I will finish the other two questions to show the full journey. Notice that submitting an answer changes real saved state; this is not a slide animation.”

Exact practical actions: LIVE APP: read question 1, invite one panel member to choose, and wait up to 30 seconds. Click the chosen option, then Submit answer. Click Next question. Select and submit question 2, then Next question. Select and submit question 3, then See my results. Submit the panel’s actual choice even if wrong. See the private answer key later in this PDF. Feedback remains hidden until completion. Allow about 75 seconds for reading, clicks, and participation. SLIDE 5 is the backup.

6:05-7:05 Show what actually changed

Slide cue: 6 / Saved progress / LIVE APPLICATION; slide is a backup

Say:

"Here is the result from the answers we just submitted. Let us return to the learning plan and then the student profile. The current estimate uses the recent response history, so it is different from the percentage correct in this single check. A good immediate result is useful evidence, but it does not establish long-term retention. A later assessment would be needed for that. What we can demonstrate now is the completed loop: a gap led to a lesson, the lesson led to a check, and the check produced a saved result that faculty can use."

Exact practical actions: LIVE APP: read the actual correct count. Click View updated learning plan to show Evidence snapshot. From the original baseline, a fully correct retest produces 20% before and 40% current, while the retest result is 3/3. Click Return to student profile. If an answer was wrong, say "This result tells us that more support is needed." Do not claim predetermined improvement. SLIDE 6 is the backup.

7:05-8:05 Show a different student need

Slide cue: 7 / Compare Asha and Rahul / LIVE APPLICATION; slide is a backup

Say:

“Now let us look at Rahul. His profile does not look like Asha's. His off-by-one and nested-loop estimates are tied at the low end; the system selects off-by-one first in the course order. An off-by-one problem means a loop may run once too few or once too many. His focused lesson therefore checks the intended first and last values, rather than repeating Asha's nested-loop explanation. This is the practical meaning of targeting support at a concept. The teacher can inspect the other concepts too, and choose a different starting point when their classroom knowledge suggests one.”

Exact practical actions: LIVE APP: click Compare students in the sidebar. Select Asha Raman and Rahul Raj if they are not already selected. Show both evidence profiles. Open Rahul’s student profile using the profile link in his comparison card, then click Open learning plan. Show the off-by-one lesson; do not complete another retest. If the displayed values have changed, describe them as shown. SLIDE 7 is the backup.

8:05-9:00 Handle missing evidence honestly

Slide cue: 8 / Ishaan and missing evidence / LIVE APPLICATION; slide is a backup

Say:

"Ishaan is a third situation. We do not yet have answers from him. The system therefore shows that he has not been assessed. It does not assign him a zero or declare that he needs urgent support simply because data is missing. His next step is a short diagnostic. After those first answers, the evidence remains limited; two responses are only a starting point. That distinction matters in a real class: a missing record and a demonstrated learning gap should lead to different conversations. Here, the teacher can see which situation they are dealing with."

Exact practical actions: LIVE APP: click Students, type Ishaan in Search name or ID, then click Ishaan Das. Point to Not assessed and missing engagement values. Click Start diagnostic and show the first question. Submission is optional. If you complete the two questions, retain the Limited evidence qualification. Allow 10 seconds to inspect the difference. SLIDE 8 is the backup.

9:00-10:00 Bring a mentor into the loop

Slide cue: 9 / Omar and mentor follow-up / LIVE APPLICATION; slide is a backup

Say:

“Finally, there is Omar. Some difficulties need a conversation with a person. The mentor-review area makes that next action visible, with the student's name, concept, and reason for review. A teacher can open the evidence and acknowledge the item. Acknowledging means it has been seen; it does not mean the learning difficulty has disappeared. It also does not send a message or email. This is the faculty workflow that connects the learning activity to follow-up. EduPulse keeps the teacher involved in deciding what kind of support the student needs.”

Exact practical actions: LIVE APP: click Mentor review. Show Omar Farooq’s existing record. Click Inspect learning evidence, then return with the sidebar Mentor review link. Click Acknowledge review if still open. If already acknowledged, select All reviews. This is a prepared case; do not claim Asha’s answers created it. After this section, return to PowerPoint with Alt+Tab and jump to SLIDE 10. SLIDE 9 is the backup.

10:00-11:15 Explain how it works

Slide cue: 10 / Architecture and the role of AI / PROJECT THE SLIDE

Say:

“The product connects learner activities with a faculty workspace. React and TypeScript render the screens, React Router connects the learning journey, and Vite builds the interface. A submitted answer goes to Python and FastAPI for validation. The server checks the reviewed answer key and saves the response in SQLite. The rules engine then recalculates concept evidence and support priority, which drive the lesson, retest, and faculty follow-up. The optional AI branch sits inside explanation delivery. When a provider is approved and configured, Gemini can use the selected reviewed lesson as context and return a structured explanation. The server validates that structure and its source identifier before using it. Grading stays with the reviewed answer keys. This architecture connects diagnosis, targeted teaching, verification, and a teacher's next action; it does not depend on a model deciding whether a student's answer is correct.”

Exact practical actions: Return to SLIDE 10. Trace learner and faculty screens, React Router UI, FastAPI validation, SQLite records, the evidence rules, lessons and retest, and faculty review. Point to Gemini as the optional provider-backed explanation branch. Discuss present integration status candidly if asked, using the private implementation note. SLIDE 12 is an optional deployment-roadmap appendix after the closing slide 11.

11:15-12:00 Close with the next step

Slide cue: 11 / Practical value and next steps / PROJECT THE SLIDE

Say:

"The result is a working connection between a class view and a student's next step. Asha receives focused support, Rahul gets a different lesson, Ishaan begins with assessment, and Omar stays connected to a mentor. My next step would be a supervised classroom pilot: review the teaching content with faculty, use consented records, and measure delayed learning as well as immediate answers. The question I want EduPulse to help a teacher answer is simple: who needs help with what, and what should we do next? Thank you. I would be happy to take your questions."

Exact practical actions: Advance to SLIDE 11 and deliver the closing statement. Stop at 12 minutes and invite questions. Keep approximately three minutes of the fifteen-minute slot available. If a question requires evidence, return to the saved app screen rather than guessing.

Private answer key

Do not project this section or read the option letters as if they were the panel's choice. Submit any panel answer honestly, even if it is wrong. The text of the option is authoritative.

Asha nested loops independent retest

Item 1

Question: Three outer iterations and four inner iterations

Correct answer: 12

Current option: C

Reason: Each of 3 outer iterations runs all 4 inner iterations.

Item 2

Question: Outer range 1 through 4 with inner range(row)

Correct answer: 10

Current option: B

Reason: Count 1 + 2 + 3 + 4 inner executions.

Item 3

Question: Rows 1 and 2 with columns 0 and 1

Correct answer: [(1, 0), (1, 1), (2, 0), (2, 1)]

Current option: A

Reason: The inner loop finishes before the next outer iteration.

With the initial Asha baseline and all three retest answers correct, the nested-loop estimate goes **20% > 40%**. The retest itself is **3/3**. These are different measures: the estimate uses 100 × (correct + 1) / (answered + 2) across the most recent eight eligible distinct observations. It is not the percent

increase in learning, an exam-mark prediction, or proof of retention.

If the panel answers incorrectly, say: “This check shows that more support is needed. That is useful evidence too. The system records the actual answers instead of assuming that the lesson worked.”

If every answer is correct, say: “All three answers were correct in this immediate check. That is encouraging evidence. A later check would still be needed to assess retention.”

Ishaan diagnostic if time permits

Item 1

Question: list(range(4))

Correct answer: [0, 1, 2, 3]

Current option: B

Item 2

Question: list(range(2, 6))

Correct answer: [2, 3, 4, 5]

Current option: C

Two correct diagnostic answers give a 75% smoothed estimate with **limited evidence**. Do not call that established mastery. The estimate is (2 + 1) / (2 + 2).

Rahul off by one retest if specifically requested

Item 1

Question: Count labels in range(1, 6)

Correct answer: 5

Current option: A

Item 2

Question: Valid nonnegative positions of a five-item list

Correct answer: [0, 1, 2, 3, 4]

Current option: D

Item 3

Question: Sum numbers in range(1, 6)

Correct answer: 15

Current option: B

This extra retest is outside the twelve-minute route. Offer it only if the panel asks and time remains.

Where AI fits in this implementation

CURRENTLY DISABLED: no cloud model calls are needed for this presentation. Keep this page private unless the panel asks about implementation.

Exact answer to give the panel

“AI has two roles in this project. I used AI-assisted development to help implement and review the system. Separately, the application has an optional Gemini integration that can rephrase a reviewed lesson. That integration is currently off. The running presentation uses the local lessons, and the scoring and answer checking are deterministic. I can show you precisely where the optional model would enter the request flow.”

The practical point where a model would be used

In an enabled configuration, the lesson page can show **Personalize explanation**. That button is intentionally hidden while cloud generation is disabled. It sends a request to **POST /api/interventions/{id}/personalize**. The implementation is the **personalize** function in **backend/app.py**.

The backend retrieves the selected concept's reviewed lesson and misconception description. It supplies that content as context and requests structured text: **title, summary, steps, takeaway, and source_id**. It tells the model not to change the code example or invent citations.

The optional provider is configured as **Gemini 3.5 Flash-Lite**. The application reserves a usage allowance, counts the input tokens, and requests structured JSON. Pydantic validates the fields and lengths. The backend checks the source identifier and completion status before storing the explanation. The original code example and answer keys are not generated by the model.

A matching source identifier and valid JSON do not prove factual accuracy. If generation fails or the returned content fails these checks, the reviewed lesson remains available. There is no automatic retry and no paid-provider fallback.

What is not an AI model here

The concept evidence estimate, support-priority weights, answer grading, retest completion status, and current lesson selection are implemented rules. They do not come from a trained model. Lesson selection uses an explicit concept identifier; this version does not use an embedding model, vector database, or unrestricted retrieval system.

Show the current setting without changing it

Open **Workspace settings** and point to **Cloud generation: Disabled**. Say: "The live journey you just saw uses no cloud model calls. Optional generation is an extension point, not a dependency for this presentation."

Do not add a key during the panel slot. Any future activation requires a dedicated verified Free Tier project with billing disabled. Token caps do not themselves guarantee a billable account is free.

One sentence if time is short

"AI helped me build the application, and an optional model can rephrase approved lessons; today that model is off, while the complete learning workflow and deterministic grading run locally."

Private production readiness and optional slide 12

The current system is a functional localhost prototype, not an institutional multi-user deployment. The optional provider integration is implemented but disabled in this configuration; a successful live call and account eligibility have not been verified for the presentation. Do not describe the provider as active.

SLIDE 12 is an optional technical appendix for questions after the closing slide 11. Real authentication, separate student and faculty roles, institutional authorization, a durable production database, HTTPS deployment, scheduled backups and restore testing, operational monitoring, provider approval, and consented data policies are planned production work. They are not represented as implemented features of the local workspace.

If asked how it becomes a real product, say: “The working local request flow is the foundation. The next deployment stage adds authenticated student and faculty access, server-side role checks, a durable managed database, HTTPS, backups, and approved provider configuration. Those production services would need a separately approved cost and security plan; I am not presenting them as already deployed.”

Three minute question period

Where is the AI?

"The optional Gemini integration can rephrase the approved lesson. It is off in this presentation. The running local path selects content by concept and uses deterministic scoring. I am not calling the local rules a trained machine-learning model."

Is this RAG?

"This version selects local teaching material by an explicit concept identifier. It does not use a vector database, embedding search, or an unrestricted retrieval pipeline. Semantic retrieval would be a later extension if the curriculum expands."

What is your contribution and how did you build it?

"I designed the learning workflow and used AI-assisted development with open-source libraries to implement it. The implementation connects the faculty view, concept evidence, targeted lessons, assessments, persistent storage, and mentor review. I can explain the request flow and the design choices."

Use this answer only after understanding the implementation. Be ready to trace a submitted answer: browser > FastAPI validation > reviewed answer key > SQLite record > eligible evidence calculation > refreshed interface. Acknowledge assistance in line with event rules.

What makes it different from a chatbot?

"The contribution is the implemented workflow around the response: evidence at concept level, a targeted next step, a separate check, saved outcomes, and faculty follow-up. Existing tutoring products already offer related features; I am not claiming that adaptive tutoring itself is new."

How accurate is the prediction?

"This is a transparent support-priority heuristic, not a validated failure predictor. The weights are inspectable. Predictive accuracy would require historical outcomes, an evaluation design, and held-out testing."

Why does 3 out of 3 become 40 percent?

“The three-question result and the longer evidence estimate serve different purposes. The estimate still includes earlier observations and smoothing, so one short check cannot immediately erase the history. The interface shows both.”

Does this improve marks by fifteen percent?

“That was a proposed target in the original pitch, not a measured outcome. The current system records immediate responses. A classroom pilot and delayed assessment are needed before claiming a learning gain.”

Can seventy real students log in now?

“This version is a local faculty-operated prototype with seventy generated profiles. It is not yet a deployed multi-user platform. A production release needs authentication, authorization, consent and data-management policies, deployment, and load testing.”

Why SQLite?

“It makes the local prototype easy to start and keeps records persistent without a database server or hosting account. A production deployment with substantial concurrent access would require a separate infrastructure decision.”

How do you prevent repeat answers inflating scores?

“The system reopens the existing scored assessment and excludes practice from independent evidence. Distinct response evidence prevents the same question from being counted repeatedly as fresh learning.”

Why is Omar already in mentor review?

“He is a prepared case showing the faculty follow-up workflow. The review was not created by Asha's answers during this presentation.”

Does acknowledgement notify Omar?

“No. It records that a faculty member has seen the review. Messaging, scheduling, and resolution are separate actions.”

How much does it cost to run today?

“The current Windows-local configuration makes no cloud model calls and needs no paid hosting or database. Optional cloud generation stays disabled unless a dedicated free-tier account is verified. Token controls are additional safeguards, not proof that a billable account is free.”

Recovery and time control

If a route is confusing, use the sidebar or **Back to student profile**. Submitted answers persist; do not start a new browser session to try to force a better result.

If the server stops, run **Start-EduPulse.cmd** and reopen the saved profile. Say: “Let me reopen the saved student record so you can inspect the result.” If an optional feature fails, return to the core workflow.

If you reach minute 8 before finishing Asha, shorten Rahul and Ishaan to one screen each, show Omar's existing review, and use a thirty-second technology explanation. Keep the final sentence and leave time for questions.

If given only 60 seconds:

“Imagine seventy students leaving a programming lesson with different gaps. EduPulse helps faculty decide who needs help with what. Asha gets a nested-loop lesson, Rahul needs an off-by-one explanation, Ishaan begins with assessment, and Omar needs a mentor conversation. Here is a live answer being saved and reflected in the student profile. The current system runs locally on Windows with reviewed teaching material and no cloud model calls. These are generated records, and learning improvement still needs a classroom study. The next step is a supervised pilot with delayed retesting. Thank you.”

Claims to avoid

- A trained model predicts failure or identifies an exact misconception from the available evidence.
- The project guarantees higher marks or a hackathon prize.
- Generated student records are real institutional records.
- Three correct answers establish permanent mastery.
- A mentor acknowledgement resolves the student's difficulty or sends a notification.
- A local workspace is a production login system.
- Every explanation is guaranteed to be factually correct.

Keep answers beside the application

Double-click **Open-Answer-Key.cmd**, or open **docs/ANSWER_KEY.html**. Choose **Compact corner view** or **Open small window**. Keep this private window on your laptop, or print it, while the classroom application is projected. It includes all 24 questions, option letters, exact outputs, and explanations.

Quick retest key: **Asha / nested loops: C, B, A. Rahul / off-by-one: A, D, B. Loop boundaries: C, A, D.** The corresponding output is also shown so you can check the actual question, not just memorize a letter.

Use **Compare students** before and after Asha's retest to show a saved state changing for one learner while the other learner's evidence stays independent. A complete second retest is optional; the twelve-minute route only opens Rahul's different lesson. If you do enter prepared responses, describe them as presentation inputs, not as observed learning gains.

Private technical reference

EduPulse is a Windows-local prototype for a CSE Year One class configured with 70 generated student profiles. The implemented route connects concept evidence, a reviewed lesson, practice, a separate retest, persistent progress, and faculty follow-up. The core application runs without cloud services or API keys.

Item 1

Technology: React 19

Main files: src/App.tsx, src/main.tsx

Purpose and reason: Interactive faculty workspace, profiles, lessons, assessments, and settings. Updates the screen from saved server responses.

Item 2

Technology: TypeScript

Main files: src/App.tsx, src/presentation.ts, tsconfig.json

Purpose and reason: Types application data and catches build-time integration errors.

Item 3

Technology: Vite

Main files: vite.config.ts, package.json

Purpose and reason: Development tooling and production asset build. In development, proxies API requests to the local backend.

Item 4

Technology: React Router

Main files: src/App.tsx

Purpose and reason: Named routes, links, browser history, and direct navigation between workspace screens.

Item 5

Technology: CSS

Main files: src/styles.css

Purpose and reason: Professional visual theme, responsive layout, typography, focus states, and presentation styling. No paid design kit is required.

Item 6

Technology: Lucide React

Main files: src/App.tsx

Purpose and reason: Consistent navigation and action icons bundled with the application.

Item 7

Technology: Python

Main files: backend/app.py

Purpose and reason: Assessment rules, evidence calculations, persistence, optional generation safeguards, and exports.

Item 8

Technology: FastAPI

Main files: backend/app.py

Purpose and reason: Local HTTP API, routing, validation integration, and built frontend delivery.

Item 9

Technology: Pydantic

Main files: backend/app.py

Purpose and reason: Validates request fields, answer ranges, and the structure of optional generated explanations.

Item 10

Technology: Uvicorn

Main files: Start-EduPulse.ps1

Purpose and reason: Runs FastAPI bound to 127.0.0.1:8765.

Item 11

Technology: SQLite

Main files: backend/app.py; database in runtime/

Purpose and reason: Stores students, attempts, sessions, learning plans, mentor reviews, and the AI usage ledger on disk.

Item 12

Technology: Original reviewed curriculum

Main files: data/curriculum.json

Purpose and reason: Three Python concepts, 24 original MCQs, answer keys, explanations, and official Python reference links.

Item 13

Technology: Optional Gemini via HTTPX

Main files: backend/app.py, .env.example

Purpose and reason: Rephrases approved lesson content when explicitly configured. Currently disabled; not required for the learning loop.

Item 14

Technology: pytest and FastAPI TestClient

Main files: tests/test_system.py

Purpose and reason: Isolated API and persistence checks without making provider requests.

Item 15

Technology: Windows PowerShell and CMD

Main files: Start-EduPulse.cmd, Start-EduPulse.ps1, Stop-EduPulse.ps1, Setup-EduPulse.ps1

Purpose and reason: Setup, health-checked startup, browser opening, local logging, and stopping the server.

Item 16

Technology: Node.js and npm

Main files: package.json, package-lock.json

Purpose and reason: Install frontend packages and run TypeScript/Vite builds. Node is needed for setup and development; the built application is served by Python.

Item 17

Technology: Prettier

Main files: package.json

Purpose and reason: Development-only source formatting so the components and styles are easier to inspect. It is not an application service.

Item 18

Technology: Git and GitHub

Main files: .gitignore, repository history

Purpose and reason: Version control and a source backup in the requested repository. GitHub does not host this running application.